

Afrocentric Collaborative Care: Supporting Context Specific Digital Health and Care

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ABSTRACT

Non-communicable diseases e.g., diabetes, cardiovascular conditions, and cancers etc, are slowly becoming silent epidemics in African countries. Most of them are closely linked with unique clinical manifestations and could go undetected until there is a complication. Management of these diseases requires long-term informal care, and this has resulted in a complex network of relationships between patients, complementary care providers, informal and formal caregivers. Unfortunately, there are not many studies investigating these relationships from a decolonized perspective with the intention of co-designing interventions that are unique to their lived experiences. This workshop aims to bring together researchers working on digital health and care projects in Africa to discuss the implications of existing collaborative care structures and cultures for technology, as well as near future Afrocentric approaches to collaborative care. We aim to highlight existing challenges in technology enabled collaborative care, differentiate them from established challenges in the Global North and identify design opportunities. This workshop will contribute to the growing attention shown to the African continent in CSCW and HCI research more broadly.

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CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in collaborative and social computing**;

KEYWORDS

Decolonising healthcare, Afrocentric design, Digital Health in Africa

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1 INTRODUCTION

As at 2017, greater than 60% of the global burden of disease was attributed to non-communicable diseases (NCDs) [22]. Some of the NCDs with high mortality rates include diabetes, cancer, and cardiovascular diseases, and a report by the World Health Organisation revealed that they were responsible for at least 37% of mortalities in Sub-Saharan Africa (SSA) [25]. In the same report, NCDs were declared to be responsible for 50% - 88% of deaths in seven African countries - these diseases claim approximately 400,000 lives per year. Due to notable shift in disease burden from communicable diseases to NCDs [11, 15, 23], a silent epidemic of chronic diseases may be a fast-approaching reality. NCDs are known to affect diverse indigenous populations and high-income earners resident in these countries are not left out [8]. Type 2 Diabetes (T2D) for example, is

a recurring challenge in SSA with approximately 24 million positive diagnoses and millions still undiagnosed [12].

Non-communicable diseases are usually chronic in nature and require long-term collaborative care [6]. As a result, healthcare systems have to adequately prepare for an increasing disease burden and the provision of adequate care. Additionally, most of these NCDs involve sporadic acute episodes and this also affects overall disease management [4]. Currently, there are therapeutic alternatives for the management of most NCDs and non-pharmacological options for lifestyle modifications, yet a significant percentage of the African population have not attained a standard quality of life [13]. This could be as a result of factors such as changing behavioural practices, economic disparities [10], poor healthcare infrastructures [3], flawed collaborative care systems [24], and the neglect of contextualization and socio-cultural practices in the design of Digital Health Interventions [5, 9]. The importance of collaborative care in chronic disease management cannot be over emphasized, and for this reason Scientific, it is timely to bring together CSCW and HCI researchers working in this context to discuss current and future collaborative technology trends.

Advances in Artificial Intelligence, Human-Computer Interaction, Telemedicine, genomics etc [18], have resulted in a variety of digital health and care DH interventions which are permeating the African continent [1]. The confidence and acceptance rate of these solutions differ per African country. For example, in SSA countries, technology adoption has been slow compared to High-Income countries (HICs) and reasons for this could include poor literacy levels, economic disparities, limited influence from the government and lack of trust for DH interventions [14, 16]. Unfortunately, there is insufficient evidence on how to support the context specific technology needs through Afrocentric Collaborative Care design, although there is growing interest in CSCW and HCI [2, 17, 19, 20].

There are significant differences between African countries and other parts of the world, and significant contrast between African countries themselves. It is important to understand the context of a target population so as to 1) Encourage Equality, Diversity and Inclusion in design practices [7, 21], 2) Facilitate the development of interventions that have considered previously ignored nuances and, 3) Increase confidence and adoption rates of DH Interventions. Thus, it can be argued that worldviews should not be imposed on a population as a 'one size fits all' approach to design is visibly flawed.

2 WORKSHOP THEMES AND OBJECTIVES

Digital Health Researchers and enthusiasts, clinicians overseeing chronic disease care, pro-Afrocentric interest groups and individuals with an interest in providing context specific care are invited to engage in this workshop. The objectives of this workshop are:

- 1) To explore the current scope of collaborative care in African countries
- 2) To discuss the impact of context and socio-cultural structures on collaborative care and the design of digital interventions
- 3) To participate in envisioning Afrocentric Collaborative Care Futures

These shall be discussed under the following themes:

- 1) **Care pathways:** This theme will consider **formal** and **informal**

care pathways, with an aim to elaborate on the complex network of collaborative care in most African countries.

- 2) **Absence and lack of access to Digital Health interventions:** We hope to expand on the different barriers impacting the design and adoption of Digital Interventions.

3 WORKSHOP STRUCTURE

3.1 Format and Call for Participation

We propose that the workshop encourages online participation. This will permit individuals unable to travel to the conference in different zones to participate, particularly from the AfriCHI community. We propose a six hour workshop, this will grant us enough time to have in-depth discussions and conduct activities without being under pressure to summarize. We will cooperate with the organisers of the CSCW conference so as to encourage participants to attend and engage in other workshops. For example, adhering to deadlines and being responsible for tools/technical equipment that may facilitate the workshop.

3.1.1 Workshop Submissions. Firstly, an announcement will be made via the organiser's networks and website providing relevant information about the workshop and links for other researchers to make submissions. A call for submissions will be advertised on the website and CSCW, HCI, and African specific health email lists to encourage participation in the hybrid workshop. Interested participants from academia will be encouraged to submit two-to-four-page submissions in ACM CHI format that report on work-in-progress, position papers, or reflections on past research on collaborative care in Africa. These papers will be assessed on their appropriateness to the workshop call, their alignment with the workshop themes and their potential to provoke discussion at the workshops. They will be juried by the organisers of the workshop. Other participants such as clinicians or Digital Health enthusiasts will be encouraged to submit reports, posters, or short 90 second videos with their tailored biographies. We are aiming for a maximum number of thirty (30) participants.

3.2 Logistics

As it is important to allow for hybrid participation to encourage involvement beyond those who can travel to CSCW, we propose that the workshop will be hosted in-person and using an accessible video-conference platform that will encourage and support engagement in the form of chats, breakout rooms and live captions e.g., Zoom. Before the workshop, we shall provide relevant information such as: the agenda of the workshop, workshop facilitators, contact emails, video conference links. Additionally, position papers and other submitted materials will be made available on the workshop's website or open access publication platforms. We will also be asking if participants would like to share their email addresses and join a mailing list to allow for a community to be formed around Afrocentric Collaborative Care.

3.3 Pre-Workshop Activities

We will be setting up a website to advertise the papers accepted to the workshop as well as communication channels. This will include Slack so that the participants can contact each other prior to the

workshop, discuss online during the hybrid workshop, and build community and connections long after the end of the workshop.

3.4 Tentative Timeline

- 10th JULY 2023 : Call for participation advertised widely
- 25th AUGUST 2023: Submission deadline for authors
- 8th SEPTEMBER 2023: Notification of acceptance
- 18th SEPTEMBER 2023: Camera-ready papers/other submissions due
- 14th OCTOBER 2023: Workshop

3.5 Tentative Workshop Schedule

- 09:00 – 09:05: Introduction by organiser
- 09:05 – 09:20: Opening Keynote
- 09:20 – 10:20: Lightning talks/Presentations A
- 10:20 – 11:00: Online break
- 11:00 – 12:15: Round table discussion of themes/Break out rooms
- 12:15 – 01:00: Panel discussion
- 01:00 – 02:00: Lightning talks/Presentations B
- 02:00 – 02:30: Question and Answer session
- 02:30 – 03:00: Closing Keynote, wrapping up and reflections

3.6 Description of Activities

- (1) **Opening Keynote:** The Keynote speaker will introduce the workshop, highlight objectives and provide a general structure of the workshop to attendees.
- (2) **Lightning talks/Presentations:** The participants will be divided into two groups based on the workshop's themes. Each group will have a maximum of 10 participants. They will be given approximately 6 minutes each to present their work.
- (3) **Introduction of themes/Break out rooms:** We propose that there will be two break out rooms to align with each theme of this workshop. These themes and related objectives will be displayed to attendees, and they will have the liberty of selecting either of them based on their interests. Attendees will discuss and share ideas in each breakout room. Thereafter, representatives from each group will present to the whole group.
- (4) **Panel discussion:** Panelists will have discussions centered around the themes
- (5) **Question and Answer session:** In this segment of the workshop, panelists will be addressing concerns from the panel discussion and answering questions.

4 EXPECTED OUTCOMES

We expect to:

- Encourage conversations between research communities and individuals from different backgrounds.
- Invite contributors to submit their work as extended papers to make up a special issue publication on Afrocentric Collaborative Care in a journal such as CSCW Journal or IJHCS.

5 CALL FOR PARTICIPATION

This workshop explores the implications of existing collaborative care structures and cultures for technology, as well as near future Afrocentric approaches to collaborative care. This workshop will be held online in conjunction with ACM CSCW 2023 in Minneapolis, MN, USA.

The importance of collaborative care in Africa cannot be over emphasized, and for this reason, it is important to bring together CSCW and HCI researchers working in this context to discuss current and future collaborative technology trends. Advances in Artificial Intelligence, Human-Computer Interaction, Telemedicine, genomics etc., have resulted in a variety of digital health and care interventions which are permeating the African continent. The African context has different implications for the success of any technology interventions, and there are significant differences between countries and regions that are often overlooked in western-focused approaches to innovation and research in CSCW and HCI. This workshop will take a decolonized perspective with the intention of exploring past, current and future research on participatory and context specific interventions that are unique to lived experiences of health and care on the African continent.

This workshop will bring together researchers working on digital health and care projects in Africa to discuss. Together we will highlight existing challenges in technology enabled collaborative care, differentiate them from established challenges in the Global North and identify design opportunities. This workshop will contribute to the growing attention shown to the African continent in CSCW and HCI research more broadly. As it is important to allow for hybrid participation to encourage involvement beyond those who can travel to CSCW, the workshop will be hosted online using an accessible video-conference platform that will encourage and support engagement in the form of chats, breakout rooms and live captions.

Interested participants from academia should make two-to-four-page submissions in ACM CHI format that report on work-in-progress, position papers, or reflections on past research on collaborative care in Africa. These papers will be assessed on their appropriateness to the workshop call, their alignment with the workshop themes and their potential to provoke discussion at the workshop. They will be juried by the organisers of the workshop. Other participants such as clinicians or Digital Health enthusiasts will be encouraged to submit reports, posters, or short 90 second videos with their tailored biographies.

All submissions should be submitted to afrocentricareworkshop@gmail.com by 25th August AOE. More information on the workshop can be found on the website: <https://afrocentricareworkshop.blogs.bristol.ac.uk> and more information on CSCW can be found here: <https://cscw.acm.org/2023/>

6 ORGANIZERS

Oritsetimeyin Arueyingho is a doctoral candidate in Digital Health at the University of Bristol. His academic backgrounds are pharmacy and healthcare management, and his research is centered on Afrocentric approaches to Type 2 Diabetes Management in Nigeria. He utilizes pragmatic mixed research methods to understand the influence of contextualization in collaborative care and how it

could impact the design of digital interventions.

Nicola J. Bidwell is a Digital Ethics Lead at Melbourne University, University Fellow at Charles Darwin University, Australia and Adjunct Professor at the International University of Management, Namibia. Her expertise includes community-based action research for technology design in the global south. She has worked with rural inhabitants of Kenya, Mozambique, Namibia, and South Africa. Her research has designed and informed successful tech initiatives and advocacy, and is sensitive to gender and local perspectives.

Anicia Peters is the Chief Executive Officer of the National Commission on Research, Science and Technology (NCRST) of Namibia. She previously held the position of Pro Vice-Chancellor for Research, Innovation and Development at the University of Namibia, and was also the Chairperson of the Namibia Presidential Task Force on the Fourth Industrial Revolution. She co-authored the Namibian e-health strategy under a consultancy with the World Health Organisation, and also developed associated e-health platforms as per the strategy.

Jacki O'Neill is the Director of Microsoft Africa Research Institute. Her research is focused on developing innovative technologies using ethnographic methods. She has made significant contributions to HCI theory, concepts and models in Africa.

Oussama Metatla is a Senior Lecturer in Human-Computer Interaction (HCI) at the University of Bristol. His academic and research vocation are to explore afrocentric approaches to design and care alongside demonstrating how HCI as an applied field of inquiry can contribute to making human society more inclusive of people with disabilities. He is also particularly interested in utilizing mixed method approaches to HCI research, combining theory with field work, co-design, and controlled studies and evaluation.

Amid Ayobi is a Digital Health Lecturer at University College London and is part of the UCL Interaction Centre. His research is centered on informing the design of supportive digital health technologies and investigating the lived experiences of people from diverse ethnic backgrounds.

Makuochi Samuel Nkwo is a lecturer in the School of Computing and Mathematical Science, at the University of Greenwich, London. He specializes in Human-Computer Interaction (HCI), Persuasive Technology, and Responsible Artificial Intelligence (AI), and he's actively interested in Data Science/Machine Learning. His research and design works center around engagements to understand user perspectives and behaviors, digital health, collaborative designs and development, social computing, sustainable community-based interventions, and environmental sustainability.

Rockefeller Zimba is the head of Integrated complex care at the NHS Northwest London Clinical Commissioning Group (NWL CCG) and clinical expert in NHS England. He has made significant input in facilitating the provision of care in the United Kingdom and Zimbabwe.

Damiete Onyema Lawrence is a Business Development Team Lead at Cinfores Ltd., Nigeria. He holds a PhD in Business Policy from Rivers State University. His background spans across Management Information Systems, Human-Computer Interaction (HCI) and Business policy. He investigated the Impact of HCI on users in Higher Educational Systems and has conducted significant research focused on achieving Sustainable Development Goals (SDGs) using afrocentric approaches.

Hadiza Ismaila is a Research Fellow at the Institute of Epidemiology and Health Care at University College London. Her background is in software engineering and Human-Computer Interaction. Her research interests lie at the intersection of HCI for development and public health. She is particularly interested in designing and evaluating digital health interventions for resource-constrained communities.

Hilda Owii is a PhD student in Social Policy at the University of Bristol. At the Perivoli Africa Research Centre, she forms part of a multi-disciplinary team to work, and build her doctoral thesis, on a research and policy engagement programme, 'Care Work and Economy Africa: toward transformative care systems and economies to harness Africa's demographic transition'. This chimes with her keen research interest long term care provision and receipt in sub-Saharan Africa, its dynamics, impacts, and interaction with culture.

Saka Abiola Monsur is a Digital Health PhD student at the University of Bristol. He has a background in Computer Science and has worked as an assistant lecturer at Nile University, Nigeria. His thesis was on a Mobile Based Digital Image Analysis for Cervical Cancer Detection. His work sought to contribute to improving the early detection of cervical cancer, especially in Sub-Saharan Africa where the high cost of testing and shortage of health workers lead to late diagnosis and poor prognosis.

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